

Technical Appendix for Anonymous Review

1 Resource and Construction Details

CliniCause provides two dataset-specific, estimator-ready causal-analysis resources: MIMIC-III (26,845 analysis records) and PhysioNet 2012 (7,993). The analysis unit is a source time-series record or ICU stay, rather than a unique person. At each source boundary, accepted identifiers are normalized to the canonical `ts_id` key; long irregular events retain minute, variable, and value, while a record-level table carries in-hospital mortality and available covariates. The two resources share an interface but are never pooled: their variable vocabularies, proxy ontologies, measurement processes, and DAG assumptions remain source specific.

Table 1: Admitted original-cohort exposure sets and source-specific semantics.

Resource	Admitted exposures	Source-specific proxy boundary
MIMIC-III	Cardiac strain; inflammation/sepsis; hepatic/coagulation dysfunction; renal dysfunction; global severity; metabolic derangement; neurologic dysfunction; respiratory failure; shock.	Hepatic and coagulation evidence are represented by a combined proxy. Chronic-burden fields are retained as baseline constructs rather than admitted exposures.
PhysioNet 2012	Cardiac injury/strain; coagulation/hematologic dysfunction; global severity; hepatic dysfunction; inflammation/sepsis burden; metabolic derangement; neurologic dysfunction; renal dysfunction; respiratory failure; shock.	Hepatic and coagulation/hematologic evidence are represented separately. Chronic baseline constructs remain outside the admitted exposure set.

At design time, structured LLM proposals covered proxy ontologies, rule families, missingness considerations, and dataset-specific DAGs. Project selection encoded chosen proposals as deterministic proxy-rule and DAG source artifacts. The LLM did not receive runtime patient rows and was not a patient-level estimator. At construction time, each source is preprocessed in its own lane; deterministic rule labels and four temporal models (STraTS, GRU, GRU-D, and TCN) produce aligned annotation exports. Each export is normalized to probability and binary fields; one deterministic rule source and four model sources are then combined by a fixed deterministic aggregation. The final resource joins the aggregate exposure layer to outcome, observed covariates, DAG/adjustment metadata, and provenance records.

Validation is layered: schema handoffs check required columns, order, value ranges, binary/probability consistency, and expected output sets; cohort handoffs reject conflicting IDs, missing/extra records, and silent shrinkage; prediction exports and aggregation enforce complete aligned voters; and provenance artifacts bind available hashes, manifests, configurations, and receipts. Archived-result evidence establishes checked aggregate outputs, but the exact historical producing revisions, configurations, and checkpoint-to-export lineage remain incomplete. Current repaired contracts are documented separately and are not claimed to have produced the archived results.

2 Complete Predictive Evaluation

The predictive task is multi-label estimation of each dataset’s deterministic proxy vector. Test loss is sample-mean multi-label binary cross-entropy. AUROC, AUPRC, and minRP are calculated per target then macro-averaged after excluding target columns with a single observed evaluation class; minRP is the maximum over thresholds of the smaller of precision and recall. The table reports selected archived held-out point metrics. It does not supply repeated run uncertainty, significance testing, split-ID lineage, or a verified checkpoint-to-export mapping.

Table 2: Complete selected archived held-out predictive results. Loss is minimized; the other metrics are maximized. Degenerate target columns are excluded from macro AUROC, AUPRC, and minRP.

Dataset	Model	Loss	AUROC	AUPRC	minRP
MIMIC-III	STRATS	0.348136	0.905411	0.869417	0.795399
	GRU	0.385757	0.881119	0.839559	0.770421
	GRU-D	0.404448	0.884277	0.841273	0.772533
	TCN	0.414456	0.866914	0.822642	0.757798
PhysioNet 2012	STRATS	0.340692	0.914761	0.877456	0.818228
	GRU	0.396819	0.914616	0.897493	0.831044
	GRU-D	0.330593	0.918478	0.905105	0.834958
	TCN	0.476513	0.899103	0.875091	0.811184

3 Complete Original-Cohort Estimator Matrix

Table 3: Complete original-cohort estimator summaries (CF-DML denotes CausalForestDML). Values are mean model-estimated CATE summaries over the analyzed samples; display precision is six decimal places.

Dataset	Exposure	CF-DML	LinearDML	CausalPFN	Sign agreement
MIMIC-III	Cardiac strain	+0.220356	+0.187634	+0.259327	all three
MIMIC-III	Inflammation/sepsis	+0.161179	+0.161334	+0.148544	all three
MIMIC-III	Hepatic/coag. dysfunction	+0.097598	+0.083017	+0.096846	all three
MIMIC-III	Renal dysfunction	+0.091290	+0.088595	+0.086972	all three
MIMIC-III	Global severity	+0.062101	+0.080153	+0.073050	all three
MIMIC-III	Respiratory failure	+0.036133	+0.039098	+0.049766	all three
MIMIC-III	Neurologic dysfunction	+0.034253	+0.031550	+0.034993	all three
MIMIC-III	Shock	+0.020994	+0.021246	+0.031645	all three
MIMIC-III	Metabolic derangement	+0.019743	+0.017989	+0.031323	all three
PhysioNet 2012	Renal dysfunction	+0.120027	+0.090622	+0.110669	all three
PhysioNet 2012	Cardiac injury/strain	+0.111831	+0.122033	+0.119413	all three
PhysioNet 2012	Global severity	+0.107988	+0.106740	+0.105039	all three
PhysioNet 2012	Hepatic dysfunction	+0.090527	+0.060184	+0.106843	all three
PhysioNet 2012	Neurologic dysfunction	+0.081636	+0.075728	+0.077049	all three
PhysioNet 2012	Metabolic derangement	+0.077704	+0.079738	+0.090734	all three
PhysioNet 2012	Inflammation/sepsis burden	+0.071800	+0.070872	+0.067480	all three
PhysioNet 2012	Respiratory failure	+0.064750	+0.062660	+0.062984	all three
PhysioNet 2012	Coag./heme dysfunction	+0.010794	+0.004136	+0.025428	all three
PhysioNet 2012	Shock	-0.013849	-0.026944	+0.004122	disagreement

4 Outcome-Downsampling Comparison

All mortality-positive records are retained while mortality-negative records are sampled under the implemented configuration. This produces a different analyzed population and outcome prevalence; therefore, the table uses direction as a robustness comparison and does not treat downsampled magnitudes as transportable estimates for the original populations. The generator verifies all 57 matched original/downsampled estimator rows and exactly two sign changes.

Table 4: Original versus outcome-downsampled estimator summaries (CF-DML, Linear, and PFN denote CausalForestDML, LinearDML, and CausalPFN). Direction compares signs only; the changed population means magnitudes are not transported between conditions.

Dataset	Exposure	Estimator	Original	Downsampled	Original sign	Direction
MIMIC-III	Cardiac strain	CF-DML	+0.220356	+0.420725	+	preserved
MIMIC-III	Cardiac strain	Linear	+0.187634	+0.403937	+	preserved
MIMIC-III	Cardiac strain	PFN	+0.259327	+0.446536	+	preserved
MIMIC-III	Global severity	CF-DML	+0.062101	+0.154682	+	preserved
MIMIC-III	Global severity	Linear	+0.080153	+0.065034	+	preserved
MIMIC-III	Global severity	PFN	+0.073050	+0.190187	+	preserved
MIMIC-III	Hepatic/coag. dysfunction	CF-DML	+0.097598	+0.192393	+	preserved
MIMIC-III	Hepatic/coag. dysfunction	Linear	+0.083017	+0.182470	+	preserved
MIMIC-III	Hepatic/coag. dysfunction	PFN	+0.096846	+0.219766	+	preserved
MIMIC-III	Inflammation/sepsis	CF-DML	+0.161179	+0.407940	+	preserved
MIMIC-III	Inflammation/sepsis	Linear	+0.161334	+0.393109	+	preserved
MIMIC-III	Inflammation/sepsis	PFN	+0.148544	+0.419575	+	preserved
MIMIC-III	Metabolic derangement	CF-DML	+0.019743	+0.028995	+	preserved
MIMIC-III	Metabolic derangement	Linear	+0.017989	+0.026342	+	preserved
MIMIC-III	Metabolic derangement	PFN	+0.031323	+0.052237	+	preserved
MIMIC-III	Neurologic dysfunction	CF-DML	+0.034253	+0.074812	+	preserved
MIMIC-III	Neurologic dysfunction	Linear	+0.031550	+0.067685	+	preserved
MIMIC-III	Neurologic dysfunction	PFN	+0.034993	+0.076865	+	preserved
MIMIC-III	Renal dysfunction	CF-DML	+0.091290	+0.180319	+	preserved
MIMIC-III	Renal dysfunction	Linear	+0.088595	+0.180638	+	preserved
MIMIC-III	Renal dysfunction	PFN	+0.086972	+0.203699	+	preserved
MIMIC-III	Respiratory failure	CF-DML	+0.036133	+0.083960	+	preserved
MIMIC-III	Respiratory failure	Linear	+0.039098	+0.090434	+	preserved
MIMIC-III	Respiratory failure	PFN	+0.049766	+0.129317	+	preserved
MIMIC-III	Shock	CF-DML	+0.020994	+0.021534	+	preserved
MIMIC-III	Shock	Linear	+0.021246	+0.013945	+	preserved
MIMIC-III	Shock	PFN	+0.031645	+0.036761	+	preserved
PhysioNet 2012	Cardiac injury/strain	CF-DML	+0.111831	+0.225488	+	preserved
PhysioNet 2012	Cardiac injury/strain	Linear	+0.122033	+0.243940	+	preserved
PhysioNet 2012	Cardiac injury/strain	PFN	+0.119413	+0.240019	+	preserved
PhysioNet 2012	Coag./heme dysfunction	CF-DML	+0.010794	+0.011005	+	preserved
PhysioNet 2012	Coag./heme dysfunction	Linear	+0.004136	-0.017406	+	changed

Dataset	Exposure	Estimator	Original	Downsampled	Original sign	Direction
PhysioNet 2012	Coag./heme dysfunction	PFN	+0.025428	+0.055540	+	preserved
PhysioNet 2012	Global severity	CF-DML	+0.107988	+0.244412	+	preserved
PhysioNet 2012	Global severity	Linear	+0.106740	+0.206430	+	preserved
PhysioNet 2012	Global severity	PFN	+0.105039	+0.314587	+	preserved
PhysioNet 2012	Hepatic dysfunction	CF-DML	+0.090527	+0.214685	+	preserved
PhysioNet 2012	Hepatic dysfunction	Linear	+0.060184	+0.147114	+	preserved
PhysioNet 2012	Hepatic dysfunction	PFN	+0.106843	+0.216485	+	preserved
PhysioNet 2012	Inflammation/sepsis burden	CF-DML	+0.071800	+0.147896	+	preserved
PhysioNet 2012	Inflammation/sepsis burden	Linear	+0.070872	+0.141951	+	preserved
PhysioNet 2012	Inflammation/sepsis burden	PFN	+0.067480	+0.150122	+	preserved
PhysioNet 2012	Metabolic derangement	CF-DML	+0.077704	+0.187367	+	preserved
PhysioNet 2012	Metabolic derangement	Linear	+0.079738	+0.192312	+	preserved
PhysioNet 2012	Metabolic derangement	PFN	+0.090734	+0.228162	+	preserved
PhysioNet 2012	Neurologic dysfunction	CF-DML	+0.081636	+0.163649	+	preserved
PhysioNet 2012	Neurologic dysfunction	Linear	+0.075728	+0.138697	+	preserved
PhysioNet 2012	Neurologic dysfunction	PFN	+0.077049	+0.182251	+	preserved
PhysioNet 2012	Renal dysfunction	CF-DML	+0.120027	+0.226767	+	preserved
PhysioNet 2012	Renal dysfunction	Linear	+0.090622	+0.192089	+	preserved
PhysioNet 2012	Renal dysfunction	PFN	+0.110669	+0.224075	+	preserved
PhysioNet 2012	Respiratory failure	CF-DML	+0.064750	+0.110970	+	preserved
PhysioNet 2012	Respiratory failure	Linear	+0.062660	+0.095604	+	preserved
PhysioNet 2012	Respiratory failure	PFN	+0.062984	+0.131635	+	preserved
PhysioNet 2012	Shock	CF-DML	-0.013849	-0.017995	-	preserved
PhysioNet 2012	Shock	Linear	-0.026944	-0.056043	-	preserved
PhysioNet 2012	Shock	PFN	+0.004122	-0.041287	+	changed

5 Matching Availability and Support

Matching is a descriptive, support-oriented, greedy one-to-one comparison over the archived binary matching representation. Successful rows report the descriptive matched-pair outcome difference. A failure indicates that no usable binary matching columns remained after preprocessing; it is not an effect of zero.

Table 5: Original-cohort matching availability. Failures are retained as failures, not zero effects.

Dataset	Exposure	Status	Pairs	Match rate	Final distance	Result / reason
MIMIC-III	Cardiac strain	success	4769	1.000	0	+0.344726
MIMIC-III	Global severity	success	7537	0.390	1	+0.074831
MIMIC-III	Hepatic/coag. dysfunction	success	6478	1.000	0	+0.133374
MIMIC-III	Inflammation/sepsis	failure	–	–	–	No binary matching columns after preprocessing
MIMIC-III	Metabolic derangement	success	5061	0.506	0	+0.036949
MIMIC-III	Neurologic dysfunction	success	6275	0.539	0	+0.026932
MIMIC-III	Renal dysfunction	success	8915	0.915	0	+0.133595
MIMIC-III	Respiratory failure	success	10998	0.694	2	+0.105656
MIMIC-III	Shock	success	10380	0.717	1	+0.038632
PhysioNet 2012	Cardiac injury/strain	success	2840	1.000	0	+0.133803
PhysioNet 2012	Coag./heme dysfunction	success	1763	0.558	0	+0.001702
PhysioNet 2012	Global severity	success	1782	0.287	1	+0.095398
PhysioNet 2012	Hepatic dysfunction	success	1133	1.000	0	+0.136805
PhysioNet 2012	Inflammation/sepsis burden	failure	–	–	–	No binary matching columns after preprocessing
PhysioNet 2012	Metabolic derangement	success	3378	0.757	2	+0.073416
PhysioNet 2012	Neurologic dysfunction	failure	–	–	–	No binary matching columns after preprocessing
PhysioNet 2012	Renal dysfunction	success	1841	1.000	0	+0.154264
PhysioNet 2012	Respiratory failure	failure	–	–	–	No binary matching columns after preprocessing
PhysioNet 2012	Shock	success	2433	0.438	1	+0.010275

6 Sensitivity and Permutation Coverage

Table 6: Diagnostic coverage and provenance. Counts are generated from checked aggregate diagnostic records, not inferred from estimator names.

Coverage family	Checked status / boundary
Estimator-native sensitivity	Archived DML records distinguish saved-training direct, recomputed/reconstructed, partial, and failed diagnostics; they are not collapsed.
CausalPFN sensitivity	No equivalent archived CausalPFN sensitivity stage; unavailable by archived-stage design.
Original-cohort sensitivity rows	CausalForestDML: FAILED=4; CausalForestDML: PARTIAL=15; LinearDML: FAILED=4; LinearDML: PARTIAL=15.
Treatment permutations	Available only for archived non-PFN estimator stages; disruption diagnostic, not identification proof.
Outcome permutations	Available only for archived non-PFN estimator stages; disruption diagnostic, not identification proof.
Original-cohort permutation rows	CausalForestDML: outcome=19; CausalForestDML: treatment=19; LinearDML: outcome=19; LinearDML: treatment=19.

7 Provenance and Reproducibility Boundaries

Checked archived artifacts support the resource counts, selected predictive metrics, original and downsampled estimator summaries, matching availability, and the recorded diagnostic status classes. They do not establish an exact historical clean-checkout rerun. The known gaps include producing source revisions, numbered causal configurations, predictive split IDs, checkpoint-to-export linkage, raw/processed-data manifests, producing environment/hardware capture, and CausalPFN package/version alignment.

Current source inspection documents stricter runtime validation gates for schemas, canonical identifiers, voters, and artifact reuse. Those current contracts must be tested and release-validated separately; this appendix makes no current test-pass claim. Any eventual anonymous package must keep source-data access restricted to authorized MIMIC-III and PhysioNet users, include only licensing-permitted derived artifacts, and undergo a final identity and metadata audit. No patient-level values are included in this appendix.